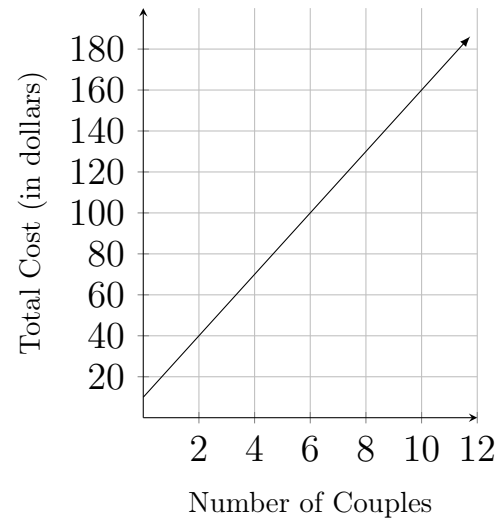


head
head

1. Answer the question below by choosing the correct response.



The graph above shows the relationship between the cost of a luncheon and the number of people attending the luncheon. According to the graph, what is the cost per couple?

- ☐ \$14.00
- ☐ \$15.00
- ☐ \$17.50
- ☐ \$20.00
- ☐ \$30.00

2. Answer the question below by choosing the correct response.

A fair coin is labeled "H" on one side and "T" on the other side. Stephen flips the coin 6 times and the result is H, H, T, H, H, H. What is the probability that the result will be a T on his seventh flip?

☐ $\frac{1}{7}$

☐ $\frac{1}{6}$

☐ $\frac{1}{3}$

☐ $\frac{1}{2}$

☐ $\frac{5}{6}$

3. Answer the question below by choosing the correct response.

Which of the following expressions is equivalent to the expression $15 - 6x$ for all values of x ?

☐ $5 + 3(5 - 2x)$

☐ $18 - (3 - 6x)$

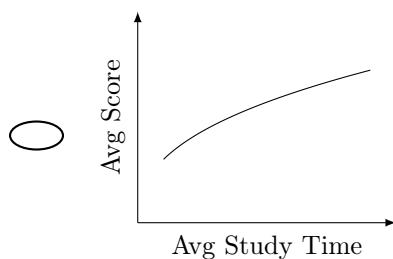
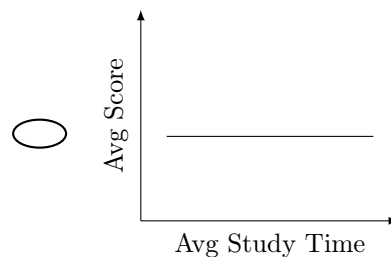
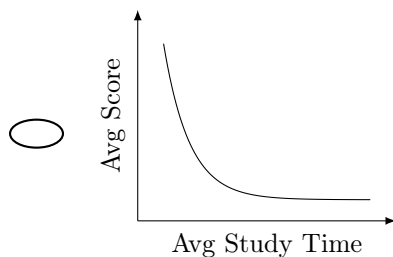
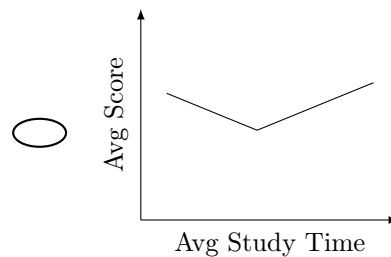
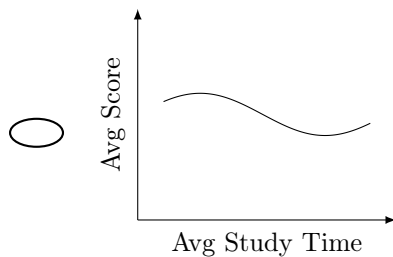
☐ $19 - 2(2 + 3x)$

☐ $3(2 - 2x) + 10$

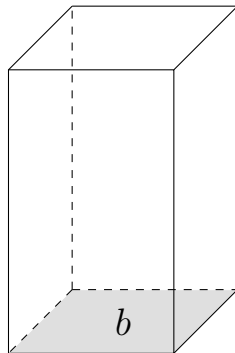
☐ $2(5 - 6x) + 25$

4. Answer the question below by choosing the correct response.

Jim observed that the more time his students spent studying for the Praxis the higher their scores were. Which of the following graphs best describes the relationship Jim observed between time students spent studying and their test scores?



5. Answer the question below by choosing the correct response.



A rectangular prism with a square base b is shown in the figure. If a plane passing through the rectangular prism is neither perpendicular nor parallel to the base b , which of the following could be the shape of the intersection of the plane and rectangular prism?

- ☐ A rectangle with a perimeter less than the perimeter of b .
- ☐ A rectangle with a perimeter equal to the perimeter of b .
- ☐ A rectangle with a perimeter greater than the perimeter of b .
- ☐ A square with a perimeter less than the perimeter of b .
- ☐ A square with a perimeter more than the perimeter of b .

6. Write your answer in the provided form.

If $-(x + 7) = -3(2x + 3) - 2$, what is the value of x ? Give your answer as a fraction.

$$x = \frac{\boxed{}}{\boxed{}}$$

7. Answer the question below by choosing the correct response.

In which of the following are the numbers ordered from least to greatest?

☐ $-\sqrt{2}, \frac{2}{5}, \frac{2}{9}, -2$

☐ $-\sqrt{2}, -2, \frac{2}{5}, \frac{2}{9}$

☐ $\frac{2}{5}, \frac{2}{9}, -2, -\sqrt{2}$

☐ $-2, -\sqrt{2}, \frac{2}{9}, \frac{2}{5}$

☐ $\frac{2}{9}, \frac{2}{5}, -\sqrt{2}, -2$

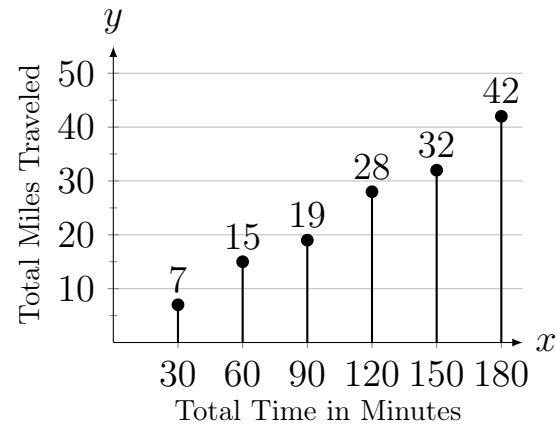
8. Indicate all of your answers.

If $\frac{2x}{7}$ is between 4 and 6, which of the following could be the value of x ?

Indicate **all** that apply.

- ☐ 7
- ☐ 11
- ☐ 15
- ☐ 19
- ☐ 23

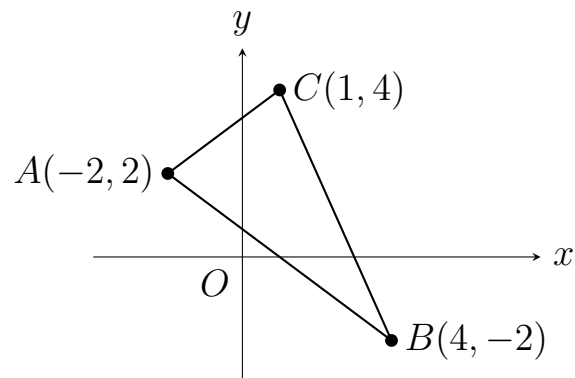
9. Answer the question below by choosing the correct response.



Will completed a 42-mile trip in 3 hours. The graph above indicates the total number of miles Will traveled at 30-minute intervals. According to the graph, how many miles did Will travel in the last 60 minutes of his trip?

- ☐ 15
- ☐ 14
- ☐ 28
- ☐ 35
- ☐ 42

10. Answer the question below by choosing the correct response.



If a triangle ABC in the xy -plane shown above is shifted 7 units to the right and 4 units up, what would be the coordinates of point A after the shift?

- ☐ $(-5, -6)$
- ☐ $(-5, -2)$
- ☐ $(-5, 6)$
- ☐ $(5, -2)$
- ☐ $(5, 6)$

11. Answer the question below by choosing the correct response.

3.4, 3.8, 4.2, 4.0, 3.0, 3.9, 3.2, 3.1, 4.7

What is the median of the numbers in the list above?

- ☐ 3.2
- ☐ 3.4
- ☐ 3.7
- ☐ 3.8
- ☐ 3.9

12. Answer the question below by choosing the correct response.

If $\frac{2}{3}x - 7 = 1$, then $x =$

☐ 8

☐ 9

☐ 12

☐ 16

☐ 24

13. Indicate all of your answers.

$$\sqrt{18}, \alpha, \frac{10}{3}, \frac{9}{8}, \frac{1}{4}, -\sqrt{2}$$

The five numbers shown are listed in order from **greatest** to **least**. Which of the following could be the value of α ?

Indicate **all** such values.

☐ $\sqrt{17}$

☐ $\frac{7}{3}$

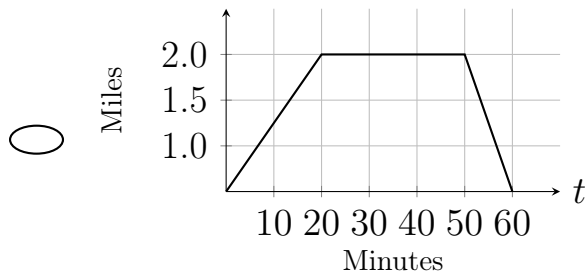
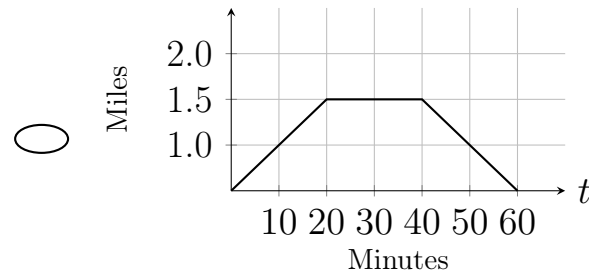
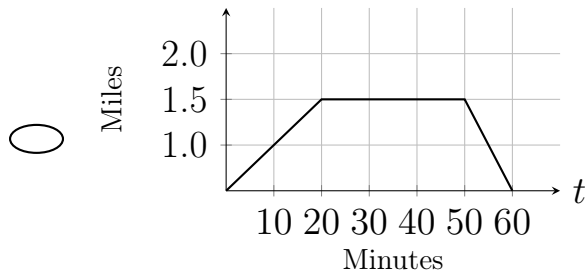
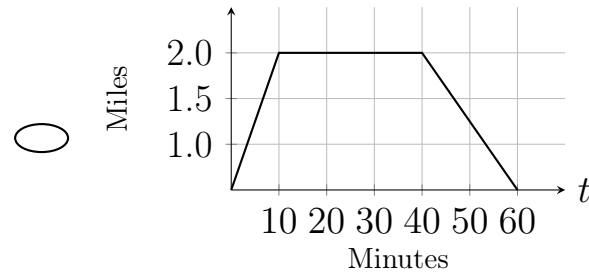
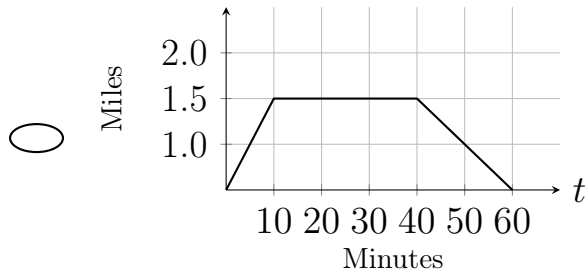
☐ $\frac{11}{3}$

☐ $\frac{23}{8}$

☐ 5

14. Answer the question below by choosing the correct response.

Last week Catherine jogged at a constant rate of 4.5 miles per hour from her work to the Happy Jack Hamburger restaurant. She stayed at the restaurant for 30 minutes and then ran back to work at a constant rate of 9 miles per hour. If the restaurant is 1.5 mile from her workplace, which of the following graphs best represents the relationship between Catherine's distance from work d , in miles, and time t , in minutes?



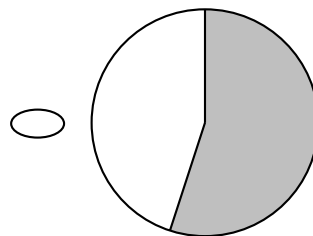
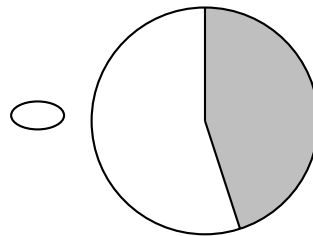
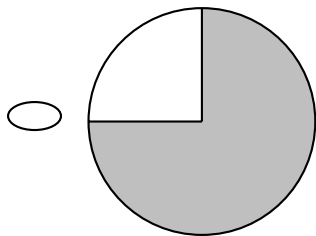
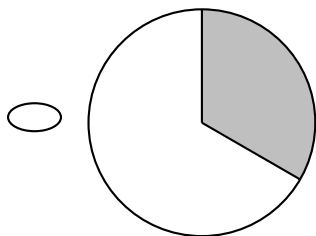
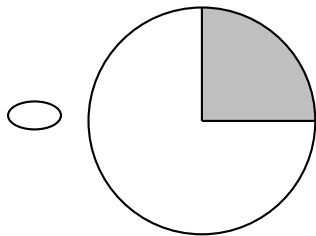
15. Answer the question below by choosing the correct response.

Which of the following sequences of steps, when completed, will solve the equation $-3 + 6z = 2$ for z ?

- ☐ Subtract 2 from both sides of the equation, then divide both sides by 3.
- ☐ Subtract 2 from both sides of the equation, then divide both sides of the new equation by -3 .
- ☐ Add 3 to both sides, then subtract 6 from both sides.
- ☐ Add 3 to both sides of the equation, then divide both sides by 6.
- ☐ Divide both sides of the equation by 6, then add 3 to both sides.

16. Answer the question below by choosing the correct response.

Last year 55 percent of the total number of sales at a gas station was for Big Time Chew bubble gum. In which of the following circle graphs does the shaded area represent the fraction of the total number of sales which were for Big Time Chew bubble gum?



17. Answer the question below by choosing the correct response.

There are 35 people on a plane, 14 are going to New York, 11 are going to Salt Lake City, and the rest are going to Asheville. If one person is randomly selected from the plane, what is the probability that the person will **not** be going to New York?

- ☐ $\frac{2}{5}$
- ☐ $\frac{3}{5}$
- ☐ $\frac{5}{7}$
- ☐ $\frac{2}{7}$
- ☐ $\frac{24}{35}$

18. Answer the question below by choosing the correct response.

The ratio of the number of teachers to the number of students on a certain field trip was 6 to 24. If the total number of people on the field trip was 120, how many teachers were on the tour?

- ☐ 20
- ☐ 24
- ☐ 30
- ☐ 480
- ☐ 600

19. Answer the question below by choosing the correct response.

Wednesday, the temperature was 60°F at 11AM and 36°F at 3PM. If the temperature fell at a constant rate from 10AM to 3PM on that day, what was the temperature at 2PM rounded up to the nearest degree?

☐ 41°F

☐ 42°F

☐ 43°F

☐ 44°F

☐ 45°F

20. Answer the question below by choosing the correct response.

$$3s + 5p$$

The expression above represents the total amount in yen, denoted ¥, earned by selling s Shopkins[©] blind bags and p Pokemon[©] balls. If a Japanese toy store has a revenue of ¥644 by selling both Shopkins[©] blind bags and Pokemon[©] balls. The amount earned by selling Shopkins[©] blind bags is what fraction of the total revenue?

☐ $\frac{s}{644}$

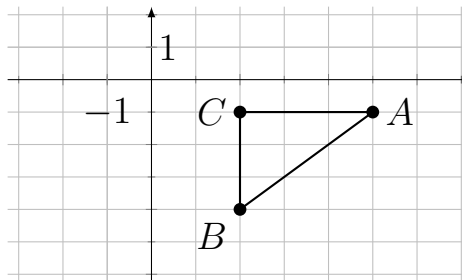
☐ $\frac{p}{644}$

☐ $\frac{3s}{644}$

☐ $\frac{5p}{644}$

☐ $\frac{3s + 5p}{644}$

21. Answer the question below by choosing the correct response.



Triangle ABC in the xy -plane above will be translated 2 units to the left and then 4 units up. What point will correspond to vertex A after these translations?

- ☐ $(-1, -1)$
- ☐ $(1, 1)$
- ☐ $(1, 5)$
- ☐ $(3, 3)$
- ☐ $(3, 5)$

22. Answer the question below by choosing the correct response.

60.5, 56.5, 56, 62, 58, 57, 55.5, 63, 59.5, 61.5, 56

Max ran the 400-meter 11 times in competitions during the spring track season. His times, in seconds, are listed above. What is the mid-range of Max's times, in seconds?

- ☐ 7
- ☐ 7.5
- ☐ 58.68
- ☐ 59
- ☐ 59.25

23. Answer the question below by choosing the correct response.

NUMBER OF PIES SOLD

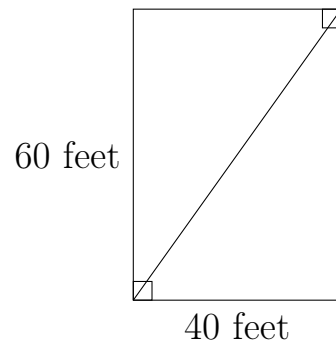
Chocolate					
Vanilla					
Carrot					

Each  represents 8 cakes.

The pictograph above shows the number of cakes sold one day at Charlie's Cake Factory. Chocolate cakes sell for \$10.50 each, vanilla cakes cost for \$10.00 each, and the total sales of the all three types of pies was \$1,012.00. How much does a carrot pie cost?

- ☐ \$10.25
- ☐ \$10.50
- ☐ \$11.00
- ☐ \$11.25
- ☐ \$11.50

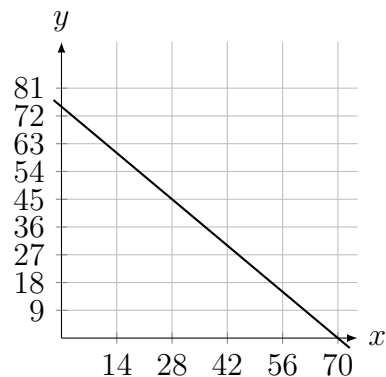
24. Answer the question below by choosing the correct response.



A farmer wants to build a diagonal fence that will divide her pig pen. The pig pen is 40 feet wide and 60 feet long, as shown in the figure. Approximately what is the length, in feet, of the diagonal brace?

- ☐ 55 feet
- ☐ 65 feet
- ☐ 75 feet
- ☐ 85 feet
- ☐ 95 feet

25. Answer the question below by choosing the correct response.



The graph of the linear equation is shown in the xy -plane above. Which of the following tables of values from linear equations rounded to the nearest tenth corresponds to the graph?

☐

x	y
28	45
42	31
56	17
70	3

☐

x	y
28	45
42	33
56	21
70	9

☐

x	y
28	42
42	26
56	10
70	-6

☐

x	y
28	44.4
42	29.6
56	14.8
70	0

☐

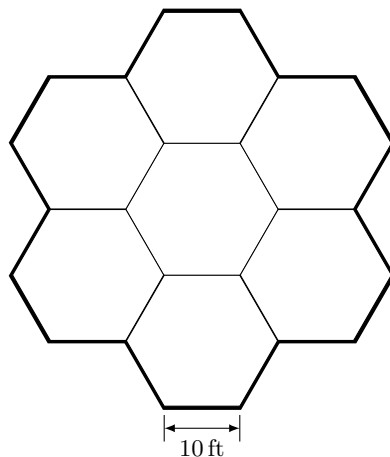
x	y
28	45
42	30
56	15
70	0

26. Answer the question below by choosing the correct response.

An administrator at a state university wants to find out how students feel about raising parking fees to pay for a new student athletic center. Which of the following methods of selecting the sample will yield the most valid information about the feelings of all the students at the university?

- ☐ Interviewing students on campus at random between classes on Monday.
- ☐ Randomly choosing forty students from the library on a given day.
- ☐ Selecting a random sample of freshmen.
- ☐ Selecting a random sample from a list of all students currently enrolled at the school.
- ☐ Sending a questionnaire to all students and using the returned questionnaires as the sample.

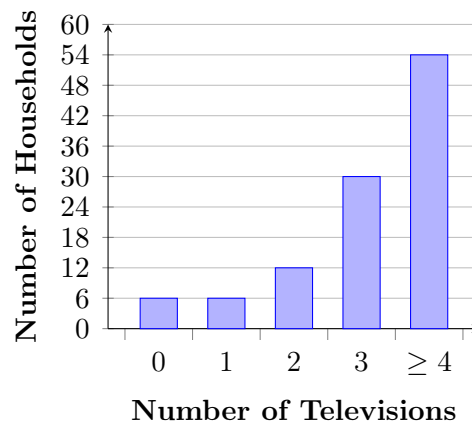
27. Answer the question below by choosing the correct response.



A large platform is constructed of 7 congruent pieces, each of which is the shape of a right prism with regular hexagon bases that have sides 10 feet long. If the platform is 5 feet tall, which of the following best approximates the volume of the entire platform? (The area of a hexagon with side length a is approximately $2.6a^2$.)

- ☐ 1,300 cubic feet
- ☐ 910 cubic feet
- ☐ 1,820 cubic feet
- ☐ 9,100 cubic feet
- ☐ 4,320 cubic feet

28. Answer the question below by choosing the correct response.



A survey of 108 randomly selected households was conducted. The graph above shows the number of households with either 0, 1, 2, 3, or 4 or more televisions. How many of the 108 households had at least 2 televisions?

- ☐ 12
- ☐ 24
- ☐ 48
- ☐ 84
- ☐ 96

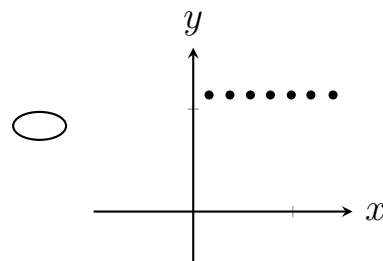
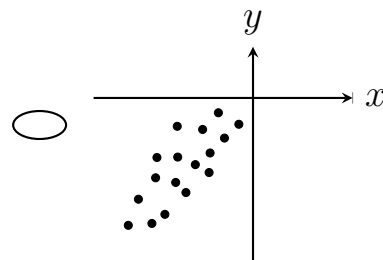
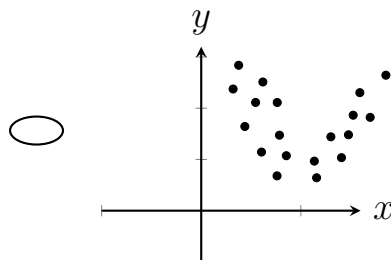
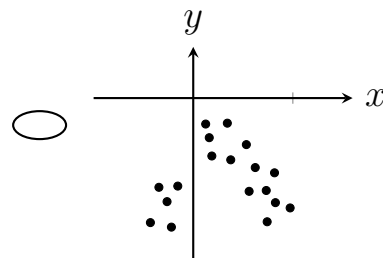
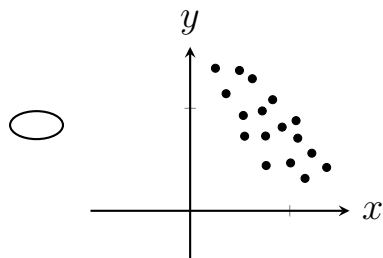
29. Write your answer in the provided form.

Alex works for an insurance firm. He earns \$1050 per month plus \$525 for each insurance policy that he sells that month. If he wants to earn more than \$6300 this month, what is the minimum number of insurance policies that he must sell this month?

insurance policies

30. Answer the question below by choosing the correct response.

Which of the following scatterplots most strongly suggests a positive linear relationship between x and y ?



31. Answer the question below by choosing the correct response.

If $a + b = 10$ and $c + d = 20$, what is the value of $(5a + 5b)(3c + 3d)$?

☐ 2,000

☐ 3,000

☐ 6,000

☐ 10,000

☐ 12,000

32. Answer the question below by choosing the correct response.

Robin sets up a conversion as follows:

$$5 \text{ miles} \times \frac{1,609,344 \text{ meters}}{1000 \text{ mile}} \times \frac{100 \text{ centimeters}}{1 \text{ meter}}$$

Which one of the following conversions is she performing?

- ☐ Miles to centimeters
- ☐ Miles to meters
- ☐ Meters to miles
- ☐ Centimeters to miles
- ☐ Miles to 1000 centimeters

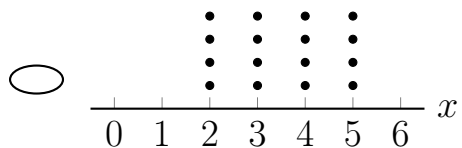
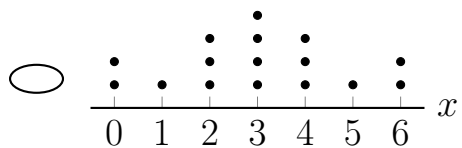
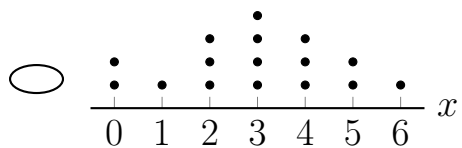
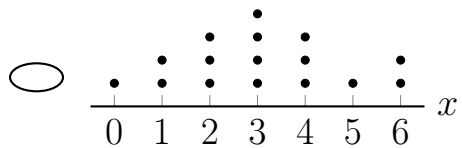
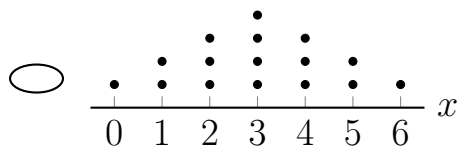
33. Answer the question below by choosing the correct response.

A painter needs to paint 4 houses, each having an area of approximately 1,450 square feet. If the brand of paint he plans to purchase covers 300 square feet per gallon, what is the minimum number of gallons he needs to purchase?

- ☐ 4
- ☐ 5
- ☐ 19
- ☐ 20
- ☐ 5,800

34. Answer the question below by choosing the correct response.

Each of the following dot plots represents a distribution of 16 values. In which distribution is the mean less than the median?



35. Answer the question below by choosing the correct response.

A is one-half as much as B , and B is 2 more than 10 times C . Which of the following statements describes the relationship between A and C ?

- ☐ A is 2 more than 20 times C .
- ☐ A is 4 more than 20 times C .
- ☐ A is 4 more than 5 times C .
- ☐ A is 1 more than 5 times C .
- ☐ A is 1 less than 5 times C .

36. Answer the question below by choosing the correct response.

A charity has \$30,000 available to spend on a fundraiser ball. The total cost t , in dollars, for the fundraiser is estimated by the equation $t = 390g + 7,800$ where g is the number of guests. What is the maximum number of guests that can attend the fundraiser ball?

- ☐ 56
- ☐ 57
- ☐ 76
- ☐ 77
- ☐ 96

37. Answer the question below by choosing the correct response.

Of the following, which is the most likely to be the height of an adult man?

- ☐ 1 meter
- ☐ 3 feet
- ☐ 132 inches
- ☐ 150 centimeters
- ☐ 2000 millimeters

38. Answer the question below by choosing the correct response.

A bag contains a number of solid-colored marbles. 13 of the marbles are blue, 5 of the marbles are yellow, and the rest are red. If a person were to randomly draw a marble from the bag, the probability that the marble drawn would be yellow is $\frac{1}{5}$. How many red marbles are in the bag?

☐ 5

☐ 6

☐ 7

☐ 8

☐ 9

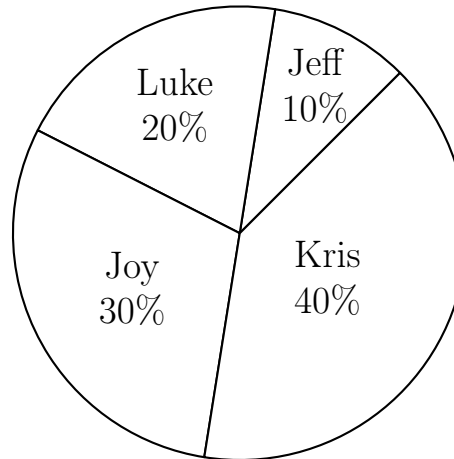
39. Answer the question below by choosing the correct response.

The price of a calculator was increased from \$90 to \$138 just before the start of school. By approximately what percent was the price of the calculator increased?

- ☐ 35%
- ☐ 48%
- ☐ 53%
- ☐ 65%
- ☐ 153%

40. Answer the question below by choosing the correct response.

DISTRIBUTION OF STAMPS
PRIOR TO TRADING SATURDAY



Kris, Jeff, Luke, and Joy collect stamps. Collectively they have a total of 550 stamps. Every Saturday they get together and trade stamps. The pie chart above shows the distribution of the 550 stamps before trading. At the end of trading Saturday night, Kris has 18 percent, Luke 30 percent, and Joy has 28 percent of the total number of stamps. How many more stamps does Jeff have at the end of trading than he had before the trading began?

- ☐ 14
- ☐ 24
- ☐ 66
- ☐ 77
- ☐ 99

41. Answer the question below by choosing the correct response.

Jim has a big wheelbarrow with a wheel 20 inches in diameter, and his son, James, has a little wheelbarrow with a wheel 10 inches in diameter. Jim pushes his big wheelbarrow straight across a field so that the wheel makes 150 revolutions. If James follows Jim and pushes his wheelbarrow the same distance, how many revolutions will the wheel of the little wheelbarrow turn?

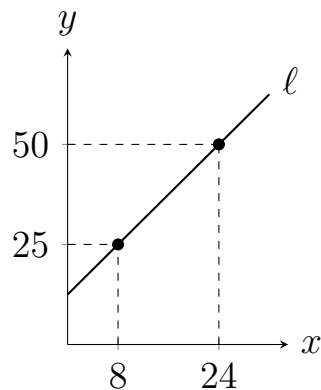
- ☐ 75
- ☐ 75π
- ☐ 150
- ☐ 300
- ☐ 300π

42. Write your answer in the provided form.

Suppose that there are five children at a birthday party. The ages of the first four children are 5, 9, 4, and 3. If the mean of the children is 6, how old is the fifth child?

$x = \boxed{}$ years old

43. Answer the question below by choosing the correct response.



The graph of line ℓ is shown in the xy -plane above. The point on line ℓ that has y -coordinate 100 is not shown. What is the x -coordinate of that point?

- ☐ 40
- ☐ 48
- ☐ 56
- ☐ 72
- ☐ 88

44. Answer the question below by choosing the correct response.

While working a math problem in class, Evaline accidentally subtracted 75.5 when she should have added 75.5. What one number can Evaline add to her final result of 1502.8 to correct the error without having to start all over?

- ☐ -151
- ☐ -75.5
- ☐ 75.5
- ☐ 151
- ☐ 1653.8

45. Answer the question below by choosing the correct response.

On a map that is drawn to scale, 8 inches represents a distance of m miles. Which of the following represents the distance, in inches, of $m + 2$ miles on the map?

☐ $\frac{8m}{m+2}$

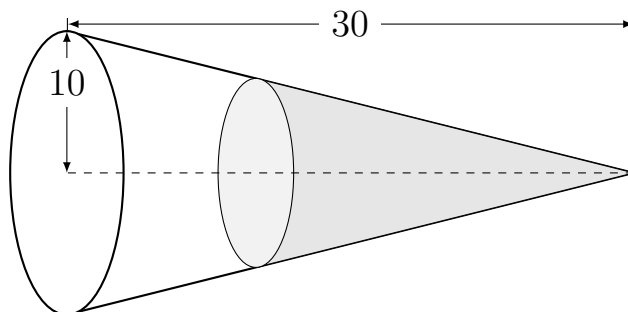
☐ $\frac{m+2}{8m}$

☐ $\frac{m(m+2)}{8}$

☐ $\frac{8}{m(m+2)}$

☐ $\frac{8(m+2)}{m}$

46. Answer the question below by choosing the correct response.



The right circular cone with a base radius of 10 centimeters and height of 30 centimeters is to be filled with lead, as shown in the figure, and fitted as the nose of a rocket. The shaded portion of the figure is also a right circular cone with a height of 0.15 meters. What is the ratio of the volume of the smaller cone to the volume of the larger cone? (The volume of a right circular cone with base radius r and height h is $\frac{1}{3}\pi r^2 h$)

47. Answer the question below by choosing the correct response.

The median and range of 117 measurements are 16 and 20, respectively. If 11 is added to each of the 117 measurements, which of the following statements regarding the median and range of the modified 117 measurements must be true?

- ☐ The median and range will both increase.
- ☐ The median and range will both stay the same.
- ☐ The median will stay the same, but the range will increase.
- ☐ The median will increase, but the range will decrease.
- ☐ The median will increase, but the range will stay the same.

48. Answer the question below by choosing the correct response.

Jill drove her mail delivery truck $10\frac{2}{3}$ miles in $\frac{4}{5}$ of an hour. What was her average speed, in miles per hour?

☐ $\frac{12}{115}$

☐ $\frac{15}{92}$

☐ $8\frac{1}{3}$

☐ $8\frac{8}{15}$

☐ $13\frac{1}{3}$

49. Indicate all of your answers.

Determine which of the following questions are statistical questions?

Indicate **all** such values.

- ☐ How tall is Michelle?
- ☐ When do high school students eat lunch?
- ☐ How heavy in grams is Jessica's calculus book?
- ☐ How many milkshakes do customers at a ice cream shop order?

50. Answer the question below by choosing the correct response.

Which of the following is equivalent to $-\frac{y}{5} + \frac{x}{2}$?

☐ $\frac{x - y}{2}$

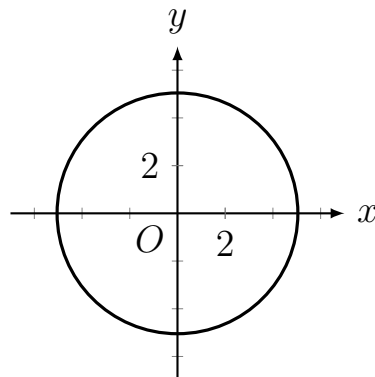
☐ $-\frac{x + y}{2}$

☐ $5x - 2y$

☐ $\frac{5x - 2y}{7}$

☐ $\frac{5x - 2y}{10}$

51. Answer the question below by choosing the correct response.



Which of the following best approximates the area of the circle shown in the graph above?

- ☐ 16
- ☐ 20
- ☐ 32
- ☐ 80
- ☐ 113

52. Indicate all of your answers.

A triangle has sides of length 5, 11, and c . Which of the following could be the value of c ?

Indicate **all** such values.

☐ 5.9

☐ 6.5

☐ 10

☐ 15.1

☐ 17

53. Answer the question below by choosing the correct response.

Jeremy planned to raise a total of x investment dollars for his startup tech business by borrowing y dollars immediately and then obtaining a fixed constant amount of dollars each week for w weeks. Which of the following expressions represents the fixed constant amount of dollars Jeremy plans to obtain each week?

☐ $x - yw$

☐ $\frac{x - w}{y}$

☐ $\frac{x}{y} - w$

☐ $\frac{x - y}{w}$

☐ $\frac{x}{w} - y$

54. Answer the question below by choosing the correct response.

If the speed of light is approximately 300,000,000 meters per second, approximately how far does light travel in 1 minute?

☐ 3.0×10^8 meters

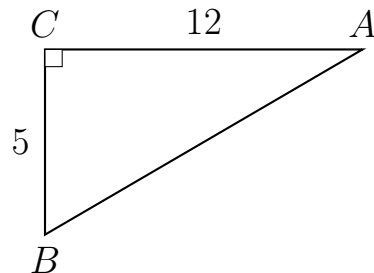
☐ 3.0×10^9 meters

☐ 1.8×10^9 meters

☐ 1.8×10^{10} meters

☐ 1.8×10^{11} meters

55. Indicate all of your answers.

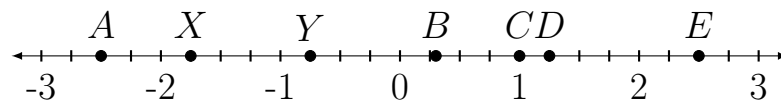


Which of the following statements must true about the triangle ABC above?

Indicate **all** such values.

- ☐ The measure of angle B is 45° .
- ☐ The perimeter of the triangle is 30.
- ☐ The sum of the measures of angle B and angle A is 100.
- ☐ The length of side AB is 17.
- ☐ The area of the triangle is 60.

56. Answer the question below by choosing the correct response.



The number line above shows 7 points, each of them labeled with a letter of the alphabet. The product of the coordinate of point X and the coordinate of point Y is closest to the coordinate of which of the following points?

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E